Statement of Work
Pompey's Pillar Iron Filter, Media, and Tank Replacement
April 2026

Background:

The proposed work includes the removal and replacement of three existing iron filtration units, replacement of exhausted filter media, rehabilitation of associated treatment components, and reactivation of the system to restore proper operation of the drinking water facility. This work is directly supported by a deficiency identified by the Montana Department of Environmental Quality (DEQ), which determined that the treatment plant is not operating as designed due to unmaintained and underperforming equipment.

The system was originally designed to remove iron through aeration followed by treatment in three pressure sand filters, with subsequent chlorination prior to entering the clearwell. Current conditions indicate that the iron and manganese removal processes are not functioning effectively. Field observations identified visible iron and substantial manganese precipitate accumulation within the storage tank and in downstream cartridge filtration located after the clearwell and prior to distribution. Analytical results further confirm treatment deficiencies, with manganese concentrations measured at 0.552 mg/L at the clearwell effluent - well above the recommended level of 0.05 mg/L - and 0.074 mg/L at the entry point to the system. Iron concentrations have been measured up to 0.10 mg/L. In addition, there is no documented record indicating that the filter vessels have ever been re-bedded, limited understanding of backwash procedures, and no available information regarding the air eductor settings necessary to achieve proper oxidation. As a result of ineffective iron and manganese removal, chlorine demand has become inconsistent, leading to variable and unreliable disinfectant residuals.

The HP-9 hydropneumatic tanks also appear to be waterlogged and are not functioning as intended. These captive air tanks are designed to maintain an appropriate air-to-water ratio to facilitate oxidation and gas release; however, both tanks exhibited a full or "waterlogged" condition when field-tested, indicating loss of air charge. Compromised hydropneumatic tanks are unable to provide adequate pressure control and may create conditions conducive to bacterial growth due to stagnant water and accumulated sediment.

Replacement of the iron filters and associated media, along with evaluation and rehabilitation or replacement of the hydropneumatic tanks, is necessary to restore proper treatment performance. Reactivation of the system will include startup procedures, verification of aeration and backwash functionality, adjustment of air eductor settings, and confirmatory sampling to ensure treatment objectives and regulatory requirements are met. This work is justified to correct DEQ-identified deficiencies, restore the system to its intended design performance, stabilize disinfection, protect public health, and ensure reliable, compliant operation of the water treatment system moving forward.

Specifications:

Tank Materials List

1. Iron Filter Vessels (Replacement)

- (3) Pressure iron filter tanks/vessels
    - Designed for potable water use
    - Sized to meet existing system flow and pressure requirements
    - NSF/ANSI 61 certified (or approved equal)
    - Construction: epoxy-lined steel or fiberglass reinforced plastic (FRP)
    - Pressure rating: typically 100–150 psi (or as required by system design)
- Interior coating (for steel tanks) suitable for corrosion protection and drinking water service

2. Hydropneumatic Tanks (Replacement)

- (2) Hydropneumatic pressure tanks (bladder or diaphragm type preferred)
    - Sized to meet system demand and minimize pump cycling
    - NSF/ANSI 61 certified for potable water systems
    - Pre-charged captive air design
    - Rated for system pressure and operating conditions
- Factory-applied internal lining suitable for potable water

General Notes

- All tanks shall be compatible with the existing treatment system and site conditions.
- Tanks shall include manufacturer certifications and pressure testing documentation.
- Connections and dimensions shall be coordinated with existing piping to minimize modifications.